

Program overview

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Year 2024/2025
Organization Architecture and the Built Environment
Education AR Electives

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Electives for non-ABE students:			
AR0095	Social Inequality in the City, Diversity and Design	5	
AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5	
AR0168	People, Movement and Public Space	5	
AR0228	Infrastructure and Environment Method Module	5	
AR8003TU	Legal and Governance	5	

Year	2024/2025
Organization	Architecture and the Built Environment
Education	AR Electives

Electives for non-ABE students:

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Instructor	Dr. R.J. Kleinhans	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and community-building, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component. Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. The acquired knowledge will be applied to a real life case study. In a Design Assignment, interdisciplinary groups of students will create strategic design proposals for a (site within a) neighbourhood which suffers from negative implications of social inequality and diversity, and for which clever solutions must be created. The design approach focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Assignment will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of (long-term) neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from social inequality and diversity, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Assignment will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Assignment sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided in advance through Brightspace.	
Reader	For the Design Assignment, a separate Design Brief will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Assignment deliverables and the final report. The summative assessment is based on two elements: 1. The Design Assignment report (60% of final grade); 2. A written exam, based on the lectures, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.8 before the final grade will be determined. If students fail to achieve the pass mark of 5.8 for the Design Assignment, they have a repair opportunity. They must resubmit an improved version of the Design Assignment report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students pass this course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. A. Meijer	
Responsible for assignments	Dr.ir. A. Straub	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0168	People, Movement and Public Space	5
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Responsible for assignments	Dr.ir. M.G.A.D. Hartevelde	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	1. Academic and grey literature review skills 2. Academic writing skills at the graduate level. 3. Basic skills are desired in either: hand drawing for reduction drawings as applicable in the urban analyses; or digital drawing as applicable in generalised image editors, such as Adobe Photoshop, or technical graphic software, like Sketchup and CityCAD.	
Course Contents	Cities are alive! People move through the city to reach their work, schools, and shops, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement, and public space - understands the main theories on people, movement, and public space - applies these theories in analysing real urban settings - evaluates critically these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer, and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, and an assignment within the scope of your discipline (track) without execution (e.g. elaborating on the design, or planning model). This will be presented/uploaded in the last week of the course. *) the discussion groups ideally consist of four/five members, who divide topics relevant to a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: ff. Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: ff. Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: ff. Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Knoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: ff. Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: ff. Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course-specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well-described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In the case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	

Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	In all cases, please understand, that you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The Faculty of Architecture and the Built Environment. (https://bis.bk.tudelft.nl/) The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. The course is an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course welcomes students from Civil Engineering and Policy Management, Complex System Engineering, Design for Social Transformations, and Infrastructure Design and Management, only if the exam regulations of your track allow this (and you follow track-specific procedures), and only if the course coordinator accepts you too. If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such to be assessed and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Course Coordinator	T. Kuzniecowa Bacchin	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge. The course programme consists of lectures, workshops and individual mentoring.	
Study Goals	Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	Prof.mr.dr. E.M. Bruggeman	
Instructor	mr. F.A.M. Hobma	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and recognize important aspects of the governance modes that develop within the institutional framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of the aspects mentioned above and to find the best possible solutions; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Ir. K.P.M. Aalbers

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